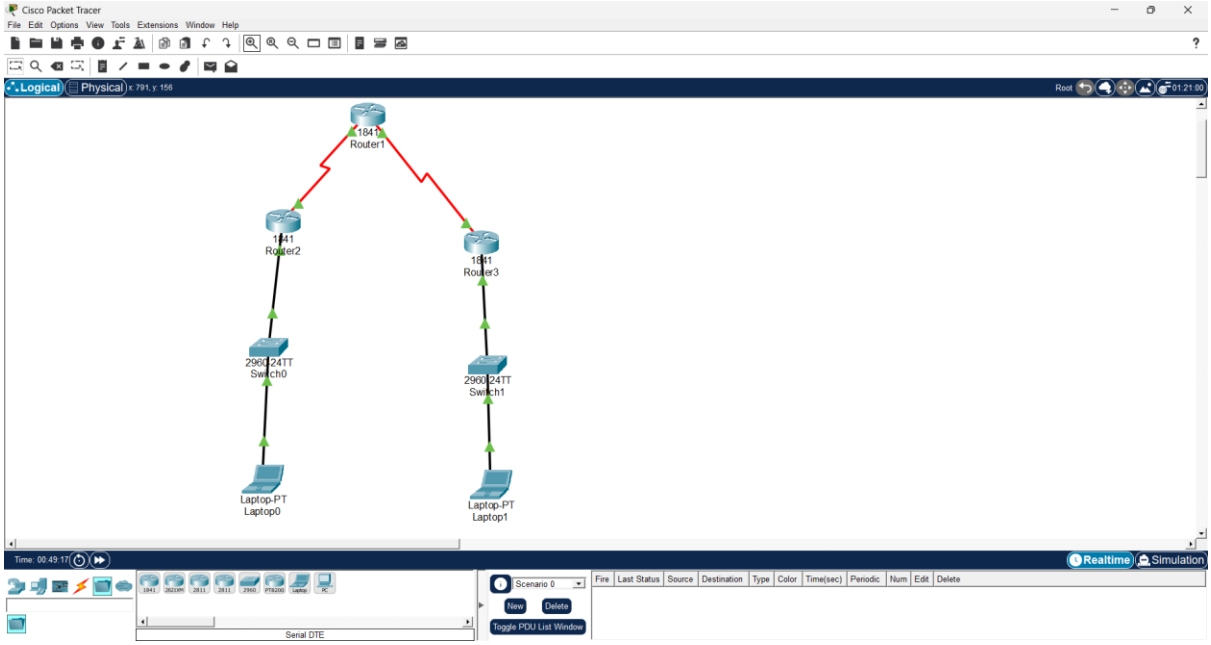
Advance Computer Network Assignment 6

Objective:

To configure and verify routing using RIP Version 2 protocol in a network.



IOS Command Line Interface

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#no ip address
Router(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#no shutdown
Router(config-if)#ip address 100.1.1.2 255.0.0.0
Router(config-if)#ip address 100.1.1.2 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#no shutdown
Router(config-if)#clock rate 56000
This command applies only to DCE interfaces
Router(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#ip address 20.1.1.1 255.0.0.0
Router(config-if)#ip address 20.1.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#router rip
Router(config-router)#version 1
Router(config-router)#no auto-summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 20.1.1.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
```

```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 20.1.1.2, 00:00:12, Serial0/0/1
     20.0.0.0/24 is subnetted, 1 subnets
C       20.1.1.0 is directly connected, Serial0/0/1
     100.0.0.0/24 is subnetted, 1 subnets
C       100.1.1.0 is directly connected, Serial0/0/0
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n


Press RETURN to get started!



Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
ip address 100.1.11.1 255.0.0.0
Router(config-if)#ip address 100.1.11.1 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 172.16.2.1 255.255.0.0
Router(config-if)#ip address 172.16.2.1 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#router rip
Router(config-router)#version 1
Router(config-router)#no auto-summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 172.16.0.0
Router(config-router)#end
Router#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available

Press RETURN to get started.

Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/2] via 100.1.1.2, 00:00:22, Serial0/0/0
R    20.0.0.0/8 [120/1] via 100.1.1.2, 00:00:22, Serial0/0/0
     100.0.0.0/16 is subnetted, 1 subnets
C       100.1.0.0 is directly connected, Serial0/0/0
C    172.16.0.0/16 is directly connected, FastEthernet0/0

Router>
```

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Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
ip address 20.1.1.0 255.0.0.0
Router(config-if)#no ip address
Router(config-if)#ip address 20.1.1.2 255.0.0.0
Router(config-if)#ip address 20.1.1.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 10.2.2.1 255.0.0.0
Router(config-if)#ip address 10.2.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#router rip
Router(config-router)#version 1
Router(config-router)#no auto-summary
Router(config-router)#network 20.1.1.0
Router(config-router)#network 10.2.2.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

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Router3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router(config-if)#ip address 20.1.1.2 255.0.0.0
Router(config-if)#ip address 20.1.1.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 10.2.2.1 255.0.0.0
Router(config-if)#ip address 10.2.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#router rip
Router(config-router)#version 1
Router(config-router)#no auto-summary
Router(config-router)#network 20.1.1.0
Router(config-router)#network 10.2.2.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
C       10.2.2.0 is directly connected, FastEthernet0/0
     20.0.0.0/24 is subnetted, 1 subnets
C       20.1.1.0 is directly connected, Serial0/0/1
R    100.0.0.0/8 [120/1] via 20.1.1.1, 00:00:11, Serial0/0/1

Router#
```

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PC1

Physical | Config | Desktop | Programming | Attributes

GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth

Global Settings

Display Name: PC1

Interfaces: FastEthernet0

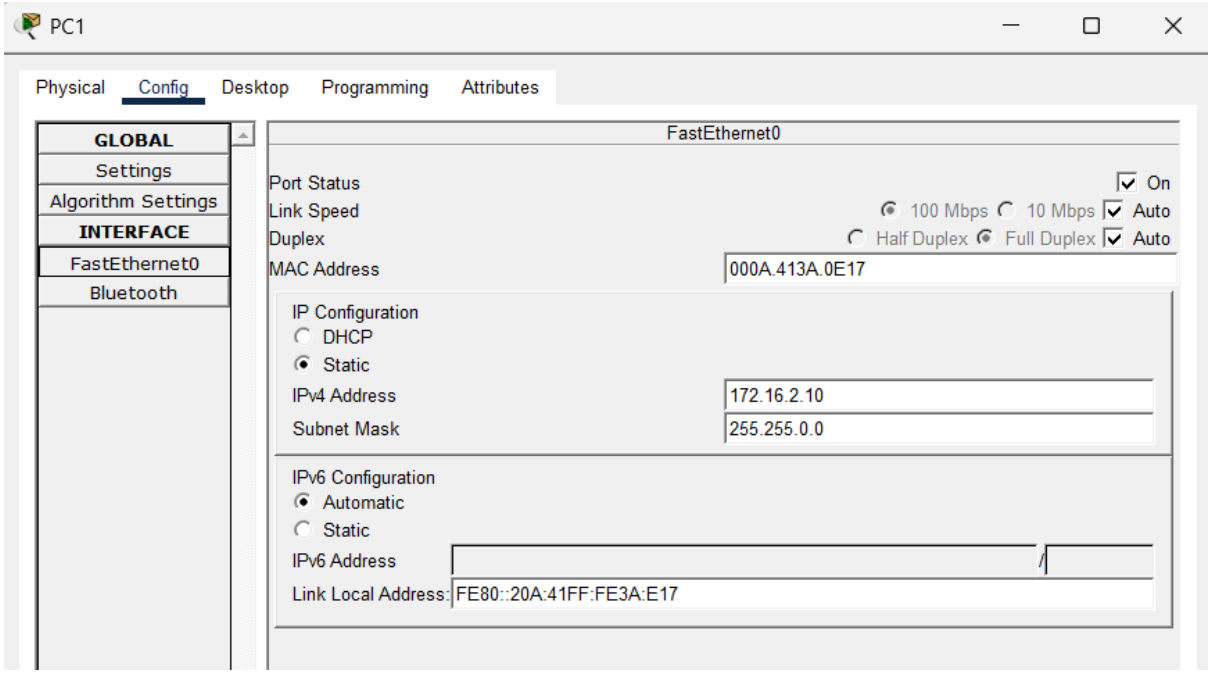
Gateway/DNS IPv4

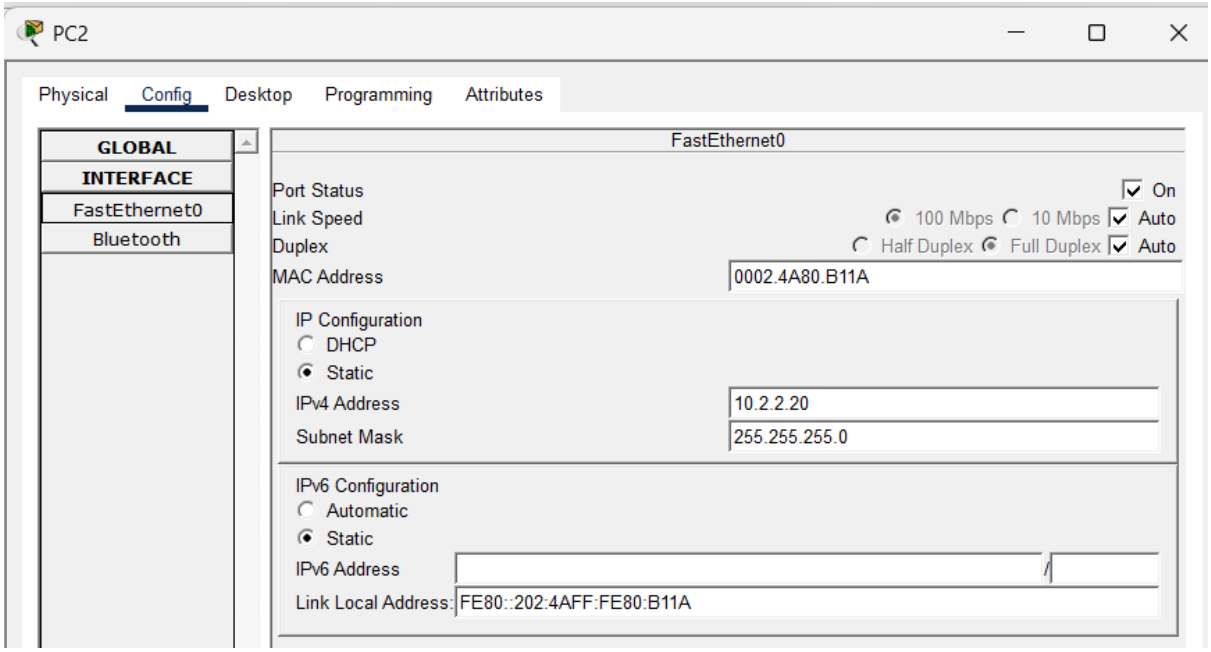
DHCP

Static

Default Gateway: 172.16.2.1

DNS Server:


PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000A.413A.0E17
IP Configuration
DHCP
Static
IPv4 Address
172.16.2.10
Subnet Mask
255.255.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::20A:41FF:FE3A:E17


PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0002.4A80.B11A
IP Configuration
DHCP
Static
IPv4 Address
10.2.2.20
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::202:4AFF:FE80:B11A